POWER PULSE

Distribution Important Formulae

Notes by Balaji Sir

1 DC distributors

1. Concentrated loads, end-fed: $\Delta V = \sum I_k\, r_k$ (loop resistance per section)

2. Uniformly loaded, end-fed: $\Delta V = \dfrac{I\,R}{2}$

I = total load current, R = total loop resistance

3. Fed at both ends: find $I_A,\ I_B$ from KVL; the point of minimum potential has $\sum I = 0$

4. Ring main: one KVL equation for the loop, then $\sum \Delta V = 0$

2 AC distribution

5. $\Delta V \approx I(R\cos\phi \pm X\sin\phi)$ (1-φ: loop R, X)

+ lagging, − leading

6. $\Delta V_L \approx \sqrt{3}\, I\,(R\cos\phi + X\sin\phi)$ (3-φ, per-phase R, X)

7. $\text{Loss} = 3I^2R = \dfrac{P^2R}{V_L^2\cos^2\phi}$

8. $V_{ph} = \dfrac{V_L}{\sqrt{3}}$ $\qquad$ 400 V → 230 V (3-φ 4-wire)

3 Economy

9. Kelvin's law: annual energy-loss cost = annual charges on the variable capital cost

gives the economical conductor area

10. $I_{economic} = \dfrac{S\ (\text{current density}) \times A}{}$ then check voltage drop and thermal limit

11. $\%\ \text{Regulation} = \dfrac{V_{NL} - V_{FL}}{V_{FL}} \times 100$

12. Ring/interconnected systems are more reliable and have lower voltage drop than radial